## 1. Introduction

### 1.1. Motivation

The interaction of a quantum system with its surroundings leads to decoherence. The standard description, the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation, rests on the Born–Markov approximations: *weak coupling* to the environment and a *memoryless* (Markovian) environment whose correlation time $\tau_E \ll \tau_S$. When these assumptions fail, as with small, structured, or strongly coupled environments, non-Markovian dynamics emerge: information flows *back* from the environment, producing effects the GKSL equation cannot capture.

Gaikwad *et al.* [1] address this by coupling a superconducting transmon qubit to a small, controllable quantum environment, using entanglement revival as an unambiguous witness of quantum memory, then tuning a dissipation parameter to drive the system through the full non-Markovian → Markovian transition.

### 1.2. Physical System and Environment

The experiment uses three **transmon qubits** fabricated at MIT Lincoln Laboratory, operated at millikelvin temperatures in the 3–5 GHz range. Each qubit is an anharmonic LC oscillator formed by a Josephson junction shunted by a large capacitor.

Table 1: Role assignment of the three transmon qubits [1].

| **Qubit** | Role and properties |
|---|---|
| **Qubit (Q)** | Frequency-tunable (SQUID loop + FFL); the open system of interest. Coupled to both A and E via parametric flux modulation. $T_2^{*(Q)} = 39\,\mu$s. |
| **Ancilla (A)** | Fixed-frequency entanglement probe; Bell partner of Q. $T_2^{*(A)} = 41\,\mu$s. |
| **Environment (E)** | Frequency-tunable small quantum bath coupled to a microwave readout resonator. Thermal photons injected into this resonator tune the dephasing rate $\gamma$ of E. |

### 1.3. Experimental Programme

The paper reports four sequential experiments, each probing a distinct aspect of the open-system dynamics:

1. **Bell-state preparation and free decay (Fig. 2).** Q and A are entangled via a $\sqrt{\text{iSWAP}}$ parametric gate, producing a Bell state with fidelity 0.91 and initial concurrence $\mathcal{C}_0 = 0.89$. With no coupling to E, the concurrence decays monotonically as $\mathcal{C}(t) = \exp(-t/T_2^{*(Q)} - t/T_2^{*(A)})$ — purely exponential Markovian behaviour confirming the GKSL baseline for the uncoupled subsystem.
2. **Non-Markovian dynamics (Fig. 3).** Q–E is parametrically coupled at rate $\Omega_{Q,E}/2\pi = 0.473\,\text{rad}/\mu\text{s}$. The Q–A concurrence collapses then *revives* at $t \sim 3$–$5\,\mu$s, violating the GKSL monotonicity. The non-Markovianity yields $\mathcal{N} = 1.4 \gg 0$.
3. **Non-Markovian → Markovian transition (Fig. 4).** Pseudo-thermal photons injected into E's resonator introduce a tunable Lindblad dephasing rate $\gamma$ on E. As $\gamma$ increases, *revivals* diminish, $\mathcal{N} \to 0$ near $\gamma \simeq 1\,\mu\text{s}^{-1}$, and fully exponential Markovian decay is recovered at $\gamma \gtrsim 3\,\mu\text{s}^{-1}$. The transition is *continuous*, not abrupt.
4. **Quantum Zeno stabilisation (Fig. 5).** In the deep Markovian limit $\gamma \gg \Omega_{Q,E}$, rapid thermal-photon measurements freeze E's state (quantum Zeno effect), suppressing Q–E coupling. *More dissipation on E preserves more Q–A entanglement*, with the decay rate approaching the natural background $\Gamma_0$.

## 2. Theoretical Framework

### 2.1. Total Hamiltonian

The total Hamiltonian separates as

$$H_{\text{tot}} = H_S + H_E + H_{SE}, \tag{1}$$

with system Hamiltonian $H_S = \frac{\hbar\omega_Q}{2}\sigma_z^Q + \frac{\hbar\omega_A}{2}\sigma_z^A$. The Q–E parametric coupling (iSWAP) is

$$H_{Q,E} = \hbar\Omega_{Q,E}\big(\sigma_+^Q\sigma_-^E + \sigma_-^Q\sigma_+^E\big). \tag{2}$$

The E–resonator dispersive interaction that mediates controllable dephasing is

$$H_{\text{int}} = \chi\, a^\dagger a\, \sigma_z^E, \qquad \chi/2\pi = 200\,\text{kHz}, \tag{3}$$

where $a^\dagger a$ is the resonator photon number operator. Fluctuating photons in the resonator produce an ac-Stark shift that dephases E at a rate calibrated empirically as

$$\gamma = 1.84\ (\mu\text{s})^{-1}\, A_{\text{in}}^{1.5}. \tag{4}$$

### 2.2. To Markovian: Derivation of the GKSL Equation

#### 2.2.1. Exact description: Nakajima–Zwanzig equation.

Tracing the total von-Neumann equation over environment degrees of freedom yields the exact integro-differential equation

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] - \int_0^t d\tau\, \mathcal{K}(t-\tau)\, \rho_S(\tau), \tag{5}$$

where $\mathcal{K}(t-\tau)$ is the memory kernel encoding all system–environment correlations. Non-zero $\mathcal{K}$ implies non-Markovian behaviour and concurrence revivals.

#### 2.2.2. Born approximation (weak coupling)

Assuming $\rho_{\text{tot}}(t) \approx \rho_S(t) \otimes \rho_E$ removes system–environment correlations. The Lindblad rates reduce to constants $\gamma_k$.

#### 2.2.3. Markov approximation (memoryless environment)

Because $\tau_E \ll \tau_S$. This eliminates the convolution in Eq. (5) and yields the time-local GKSL master equation:

$$\boxed{\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_k \gamma_k \Big( L_k\, \rho_S\, L_k^\dagger - \tfrac{1}{2}\big\{L_k^\dagger L_k, \rho_S\big\}\Big)} \tag{6}$$

The solutions of Eq. (6) are *contractive*: coherences and *concurrence can only decrease monotonically.* Any increase proves non-Markovianity.

### 2.3. Concurrence and the Non-Markovianity Measure

For a two-qubit density matrix $\rho$, concurrence is given by the Wootters formula

$$\mathcal{C}(\rho) = \max\big(0,\ \sqrt{\lambda_1} - \sqrt{\lambda_2} - \sqrt{\lambda_3} - \sqrt{\lambda_4}\big), \tag{7}$$

where $\lambda_1 \geq \cdots \geq \lambda_4 \geq 0$ are eigenvalues of the matrix $R = \rho\,\tilde{\rho}$, with spin-flipped state $\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y)$. Under pure dephasing a Bell state gives $\mathcal{C}(t) = e^{-t/T_2}$.

The Rivas–Huelga–Plenio non-Markovianity quantifies the total entanglement recovered during all revival events:

$$\mathcal{N} = \int_{t_0}^{t_f} \left|\frac{d\mathcal{C}}{dt}\right| dt\ -\ \Delta\mathcal{C}, \qquad \Delta\mathcal{C} = \mathcal{C}(t_0) - \mathcal{C}(t_f). \tag{8}$$

Subtracting $\Delta\mathcal{C}$ (total net decrease) isolates the *increasing* portions only; any $\mathcal{N} > 0$ unambiguously signals a non-CPTP map and hence non-Markovianity.

*2.4. Quantum Zeno Effect and Entanglement Stabilisation*

The survival probability: $N$ measurements in interval $T$ is

$$P_0(T) = \left[1 - \frac{(\Delta H)^2 T^2}{\hbar^2 N^2}\right]^N \xrightarrow{N\to\infty} 1, \tag{9}$$

the frequent measurements reset the state before appreciable evolution. In the experiment, the thermal-photon jumps at rate $\gamma \gg \Omega_{Q,E}$ continuously conserving the entangled state. The resulting concurrence decay rate follows the Zeno scaling:

$$\Gamma_c = \frac{\Omega_{Q,E}^2}{4\gamma} + \Gamma_0, \quad \Gamma_0 = \frac{1}{T_2^{*(Q)}} + \frac{1}{T_2^{*(A)}}. \tag{10}$$

As $\gamma \to \infty$, $\Gamma_c \to \Gamma_0$ and E is effectively decoupled from Q by its own dissipation.

## 3. Methodology

*3.1. Device Parameters*

Table 2: Key experimental parameters [1].

| Parameter | Value |
|---|---|
| $T_2^{*(Q)}$ | $39\,\mu$s |
| $T_2^{*(A)}$ | $41\,\mu$s |
| $\Omega_{Q,A}/2\pi$ | 0.477 MHz |
| $\Omega_{Q,E}/2\pi$ | $0.473\,\mathrm{rad}/\mu$s |
| $\chi/2\pi$ | 200 kHz (E–resonator coupling) |
| Initial concurrence | $\mathcal{C}_0 = 0.89$; Bell fidelity 0.91 |
| Readout fidelity | 0.97 (Q), 0.96 (A) |
| Noise bandwidth | 1.8 MHz (exceeds $\chi$; ensures Markovian dephasing on E) |
| Noise correlation time | 90 ns $\ll$ all dynamical timescales |

*3.2. Step-by-Step Protocols*

*3.2.1. Exp. 1 — Bell-State Preparation (Markovian Baseline)*

1. Apply a $\pi$ pulse to Q (microwave drive) to prepare $|e\rangle_Q$.
2. Activate parametric modulation on Q's FFL at $f_{\mathrm{mod}} = \Delta_{Q,A}/2$ for 530 ns, implementing a $\sqrt{\mathrm{iSWAP}}$ gate and producing $|\Phi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + e^{i\phi}|01\rangle)$.
3. Measure 9 Pauli pair expectation values $\{\langle\Sigma^Q\Sigma^A\rangle\}$, $\Sigma \in \{X, Y, Z\}$, via simultaneous dispersive readout.
4. Reconstruct $\rho_{QA}$ by maximum-likelihood estimation; compute $\mathcal{C}$ from Eq. (7).
5. *Result:* $\mathcal{C}(t) = e^{-t/T_2^{*(Q)} - t/T_2^{*(A)}}$ — monotonic exponential Markovian decay; validates GKSL baseline.

*3.2.2. Exp. 2 — Non-Markovian Dynamics ($\gamma = 0$)*

1. Prepare Q–A Bell state as above.
2. Activate Q–E parametric coupling by modulating both Q and E FFLs at $\Delta_{Q,E}/4 = 2\pi \times 175\,\mathrm{rad}/\mu$s.
3. Perform Q–A state tomography at each $\tau$; compute $\mathcal{C}(\tau)$.
4. Apply Eq. (8) over $[t_0{=}0, t_f{=}10\,\mu\mathrm{s}]$.
5. *Result:* Concurrence collapses then *revives* at $t \sim 3$–$5\,\mu$s, yielding $\mathcal{N} = 1.4$. This is categorically impossible under GKSL, proving quantum memory in E.

*3.2.3. Exp. 3 — Non-Markovian→Markovian Transition*

1. Drive E's readout resonator with noisy microwave tone of amplitude $A_{\mathrm{in}}$, ensuring Markovian dephasing on E.
2. Calibrate $\gamma$ for each $A_{\mathrm{in}}$ via Ramsey interferometry on E, confirming $\gamma = 1.84\ (\mu\mathrm{s})^{-1} A_{\mathrm{in}}^{1.5}$ (Eq. (4)).
3. Re-calibrate Q–E parametric coupling at each $A_{\mathrm{in}}$.
4. Repeat Exp. 2 protocol and compute $\mathcal{N}(\gamma)$.
5. *Result:* Table 3 summarises the three identified dynamical regimes. $\mathcal{N} \to 0$ at $\gamma \simeq 1\,\mu\mathrm{s}^{-1}$; Markovian behaviour (exponential $\mathcal{C}$ decay, GKSL applicable) fully established at $\gamma \gtrsim 3\,\mu\mathrm{s}^{-1}$. The transition is *continuous*: at intermediate $\gamma$, neither clear revivals nor pure exponential decay are observed.

Table 3: Three dynamical regimes identified in Exp. 3.

| **Regime** | Condition | Concurrence behaviour and physics |
|---|---|---|
| **Non-Markovian** | $\gamma{=}0$ | Collapses then *revives*; $\mathcal{N} = 1.4$. E stores quantum memory; information backflow to Q. Nakajima–Zwanzig (Eq. (5)) required. |
| **Transition** | $0{<}\gamma{\lesssim}3$ | Monotonically decaying but faster than natural rate; $\mathcal{N} \to 0$. Memory is partially destroyed; neither GKSL nor purely coherent. |
| **Markovian / Zeno** | $\gamma{\gg}\Omega_{Q,E}$ | Exponential at rate $\Gamma_c \to \Gamma_0$; GKSL fully applicable. Zeno effect (Eq. (10)) pins E, suppressing Q–E coupling and *protecting* Q–A entanglement. |

*3.2.4. Exp. 4 — Quantum Zeno Stabilisation*

1. Operate at large $\gamma$ ($\gamma{\gg}\Omega_{Q,E}$; up to $\gamma{=}45.5\,\mu\mathrm{s}^{-1}$).
2. Fit Q–A concurrence envelope to an exponential; extract effective decay rate $\Gamma_c$.
3. Plot $\Gamma_c$ vs $\gamma$ and fit with Eq. (10).
4. *Result:* $\Gamma_c$ follows the Zeno scaling $\Omega_{Q,E}^2/(4\gamma) + \Gamma_0$ to high accuracy. At $\gamma = 45.5\,\mu\mathrm{s}^{-1}$, $\Gamma_c \to \Gamma_0$: Q–A entanglement decays at the natural (uncoupled) rate by the presence of the environment.

## 4. Final Summary

This report summarises the experimental and theoretical findings of Gaikwad *et al.*, who use a superconducting transmon qubit processor to probe the transition from non-Markovian to Markovian open quantum system dynamics. Using entanglement (concurrence) as a witness, they observe collapse and revival of qubit-pair entanglement as a direct signature of quantum memory in a small engineered environment. By introducing tunable Lindblad dephasing $\gamma$ on the environment they demonstrate a continuous, controllable transition to Markovian behaviour. At high $\gamma$ the quantum Zeno effect paradoxically stabilises entanglement, establishing dissipation as a resource for quantum subspace engineering.

## References


[1] Gaikwad, C., Kowsari, D., Brame, C., Song, X., Zhang, H., Esposito, M., Ranadive, A., Cappelli, G., Roch, N., Levenson-Falk, E. M., & Murch, K. W. (2024)